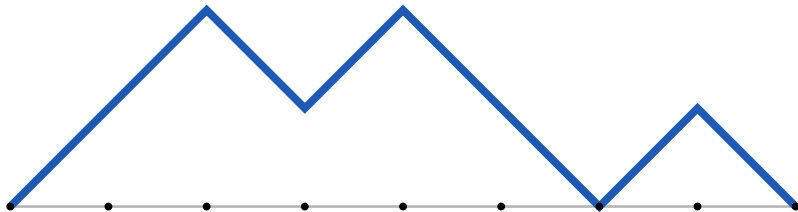


Recurrences and Catalan Numbers

Section 2.6.4–2.6.6

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. Fibonacci Numbers

2. Recurrence Relations

3. Catalan Numbers

4. Examples and Exercises

Section 2.6.4: Fibonacci numbers

Fibonacci sequence

Set $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2} \quad (n \geq 2).$$

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

A recurrence specifies each term using earlier terms, together with initial conditions.

A tiling interpretation

Question

How many ways can a board of length n be tiled using tiles of lengths 1 and 2?

Let the answer be t_n . The final tile has length 1 or 2, leaving a board of length $n - 1$ or $n - 2$.

$$t_n = t_{n-1} + t_{n-2}, \quad t_0 = t_1 = 1.$$

Tilings and Fibonacci numbers

Observation

$$t_n = F_{n+1}.$$

Proof

The two sequences satisfy the same recurrence and have the same initial values after the index shift.

Example

For $n = 4$, the tile-length words are 1111, 112, 121, 211, 22. Thus $t_4 = 5 = F_5$.

The Fibonacci generating function

Let

$$F(x) = \sum_{n \geq 0} F_n x^n.$$

Multiply $F_n = F_{n-1} + F_{n-2}$ by x^n and sum over $n \geq 2$:

$$F(x) - x = xF(x) + x^2F(x).$$

The subtracted term x accounts for $F_1 = 1$.

Solving for the generating function

Formula

$$F(x) = \frac{x}{1 - x - x^2}.$$

For the tiling numbers,

$$T(x) = \sum_{n \geq 0} t_n x^n = \frac{1}{1 - x - x^2}.$$

The characteristic roots

The equation $r^2 = r + 1$ has roots

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Since $\phi + \psi = 1$ and $\phi\psi = -1$,

$$1 - x - x^2 = (1 - \phi x)(1 - \psi x).$$

Partial fractions

$$\frac{x}{1-x-x^2} = \frac{1}{\sqrt{5}} \left(\frac{1}{1-\phi x} - \frac{1}{1-\psi x} \right).$$

Expand each denominator geometrically and compare coefficients.

Binet's formula

Theorem

For $n \geq 0$,

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}.$$

Although the expression contains irrational numbers, it is an integer because it equals the recursively defined Fibonacci number.

Growth of the Fibonacci numbers

Since $|\psi| < 1 < \phi$,

$$F_n \sim \frac{\phi^n}{\sqrt{5}}.$$

In particular,

$$\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n} = \phi.$$

The larger characteristic root determines the leading growth.

A Fibonacci sum

Identity

For $n \geq 0$,

$$\sum_{j=0}^n F_j = F_{n+2} - 1.$$

Proof

Rewrite the recurrence as $F_j = F_{j+2} - F_{j+1}$ and sum. The right-hand side telescopes to $F_{n+2} - F_1$.

A sum of squares

Identity

$$\sum_{j=0}^n F_j^2 = F_n F_{n+1}.$$

Proof

The statement holds at $n = 0$. If it holds at $n - 1$, add F_n^2 :

$$F_{n-1}F_n + F_n^2 = F_n(F_{n-1} + F_n) = F_n F_{n+1}.$$

Section 2.6.5: Recurrence relations

Linear homogeneous recurrence

A recurrence of order d with constant coefficients has the form

$$a_n = c_1 a_{n-1} + \cdots + c_d a_{n-d}.$$

The first d values determine the sequence. We begin with order 2.

The characteristic equation

For

$$a_n = ua_{n-1} + va_{n-2},$$

try a solution $a_n = r^n$. Substitution leads to

$$r^2 - ur - v = 0.$$

When the roots r_1, r_2 are distinct, the general solution is

$$a_n = Ar_1^n + Br_2^n.$$

Why two constants suffice

Proof

Each root gives a solution, and linear combinations of solutions are solutions. Distinct roots let us solve uniquely for A, B from a_0, a_1 .

The resulting sequence agrees with the required sequence in its first two terms and satisfies the same recurrence. Induction makes the two sequences identical.

An example with distinct roots

Question

Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 2$, $a_1 = 5$.

The characteristic polynomial factors as

$$r^2 - 5r + 6 = (r - 2)(r - 3).$$

So $a_n = A2^n + B3^n$.

Determining the constants

$$A + B = 2, \quad 2A + 3B = 5.$$

Hence $A = B = 1$, and

$$a_n = 2^n + 3^n.$$

Check: $a_2 = 13 = 5 \cdot 5 - 6 \cdot 2$.

Repeated roots

Order-two repeated-root case

If the characteristic polynomial is $(r - \lambda)^2$ with $\lambda \neq 0$, then

$$a_n = (A + Bn)\lambda^n.$$

The extra factor n supplies a second independent solution.

For a root of multiplicity m , the corresponding terms involve $n^j \lambda^n$ for $0 \leq j < m$.

A repeated-root example

Question

Solve $a_n = 4a_{n-1} - 4a_{n-2}$ with $a_0 = 1$, $a_1 = 4$.

The polynomial is $(r - 2)^2$, so $a_n = (A + Bn)2^n$.

Initial values give $A = 1$ and $2(1 + B) = 4$. Therefore

$$a_n = (n + 1)2^n.$$

Generating functions for recurrences

Order-two formula

If $a_n = ua_{n-1} + va_{n-2}$ for $n \geq 2$, then

$$A(x) = \frac{a_0 + (a_1 - ua_0)x}{1 - ux - vx^2}.$$

Proof

Sum the recurrence after multiplying by x^n :

$$A - a_0 - a_1x = ux(A - a_0) + vx^2A.$$

Rearrange.

The repeated-root example revisited

$$A(x) = \frac{1 + (4 - 4)x}{1 - 4x + 4x^2} = \frac{1}{(1 - 2x)^2}.$$

Using $(1 - z)^{-2} = \sum_{n \geq 0} (n + 1)z^n$ gives

$$a_n = (n + 1)2^n.$$

Partial fractions explain why repeated roots produce polynomial factors in n .

Nonhomogeneous recurrences

Definition

A recurrence is nonhomogeneous when an additional term depends on n :

$$a_n = c_1 a_{n-1} + \cdots + c_d a_{n-d} + b_n.$$

One method is to find a particular solution and add a general solution of the associated homogeneous recurrence.

A constant forcing term

Question

Solve $a_n = 2a_{n-1} + 1$, $a_0 = 0$.

Set $b_n = a_n + 1$. Then $b_n = 2b_{n-1}$ and $b_0 = 1$, giving

$$a_n = 2^n - 1.$$

A resonant forcing term

Question

Solve $a_n = 2a_{n-1} + 2^n$, $a_0 = 0$.

Divide by 2^n and set $b_n = a_n/2^n$:

$$b_n = b_{n-1} + 1, \quad b_0 = 0.$$

Thus $b_n = n$ and $a_n = n2^n$.

Recurrences from counting

Question

Let a_n count binary words of length n with no consecutive ones. Find a recurrence.

A word ends in 0, or ends in 01 if its length is at least 2.

$$a_n = a_{n-1} + a_{n-2}, \quad a_0 = 1, \quad a_1 = 2.$$

Thus $a_n = F_{n+2}$.

Section 2.6.6: Catalan numbers

Question

How many strings of n opening and n closing parentheses are correctly matched?

At every initial segment, the number of opening parentheses must be at least the number of closing parentheses.

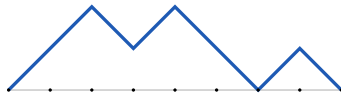
For $n = 3$, there are five:

$((()))$, $(()())$, $((())())$, $()(())$, $()()()$.

Dyck paths

Dyck path

A Dyck path of semilength n uses n up steps $(1, 1)$ and n down steps $(1, -1)$ and never goes below height zero.



Opening parentheses are up steps; closing parentheses are down steps.

The Catalan sequence

Let C_n count Dyck paths of semilength n , with $C_0 = 1$ for the empty path.

$$C_0, C_1, C_2, C_3, C_4, C_5 = 1, 1, 2, 5, 14, 42.$$

Decomposing a path at its first return to height zero gives a recurrence.

The first-return decomposition

Every nonempty Dyck path has a unique form

$$UPDQ,$$

where the displayed D is the first return to height zero, and P, Q are Dyck paths after shifting P down by one unit.

If P has semilength i , then Q has semilength $n - 1 - i$.

The Catalan recurrence

Recurrence

For $n \geq 1$,

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}, \quad C_0 = 1.$$

Example

$$C_4 = C_0 C_3 + C_1 C_2 + C_2 C_1 + C_3 C_0 = 5 + 2 + 2 + 5 = 14.$$

The Catalan generating function

Let $C(x) = \sum_{n \geq 0} C_n x^n$. The recurrence becomes

$$C(x) - 1 = xC(x)^2.$$

Solving the quadratic and choosing the branch with constant term 1 gives

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

A closed formula

Catalan formula

For $n \geq 0$,

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

We prove this directly by subtracting paths that fall below the axis from all paths with n up and n down steps.

Reflecting a bad path

Proof

There are $\binom{2n}{n}$ unrestricted paths from height 0 to height 0.

For a bad path, reflect its initial segment through the first step reaching height -1 , interchanging up and down steps. That segment had one more down step than up step; after reflection the whole path has $n + 1$ up and $n - 1$ down steps.

The reflection bijection

Proof

Conversely, a path with $n + 1$ up and $n - 1$ down steps ends at height 2 and has a first visit to height 1. Reflect its initial segment through that visit. This reverses the construction.

Therefore bad paths number $\binom{2n}{n-1}$, and

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}.$$

Triangulations of a polygon

Catalan interpretation

A convex polygon with $n + 2$ vertices has C_n triangulations.

Proof

Fix one boundary edge. The triangle incident with it has a unique third vertex, splitting the remaining polygon into two smaller polygons. Summing over that vertex gives the Catalan recurrence, with the empty region counted as 1.

Other Catalan families

- Correctly matched parentheses with n pairs.
- Dyck paths of semilength n .
- Triangulations of a convex $(n + 2)$ -gon.
- Noncrossing perfect matchings of $2n$ points on a circle.
- Rooted ordered full binary trees with n internal vertices.

Each family admits the same decomposition into a pair of smaller objects.

Question

How many correctly matched parenthesis strings with four pairs have their first return to balance zero after the fourth symbol?

In $UPDQ$, determine the semilengths of P and Q .

The initial block UPD has two pairs, so P has one pair. The remainder Q has two pairs.

$$C_1 C_2 = 1 \cdot 2 = 2.$$

The strings are $((()))((()))$ and $((()))()()$.

- Initial conditions and a recurrence determine a sequence.
- Characteristic roots and generating functions solve constant-coefficient linear recurrences.
- Repeated roots create polynomial factors.
- Catalan objects split recursively into two smaller objects and satisfy a convolution recurrence.

Permutation Groups and Burnside's Lemma